Project: Worked examples
Title: Motor and actuator sizing
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1. Scope and method

Calculates the torque, speed and power a motor needs to move a load through a mechanism, and checks a candidate motor against them. A continuously running rotary load is sized from its steady torque, speed and run-up time.

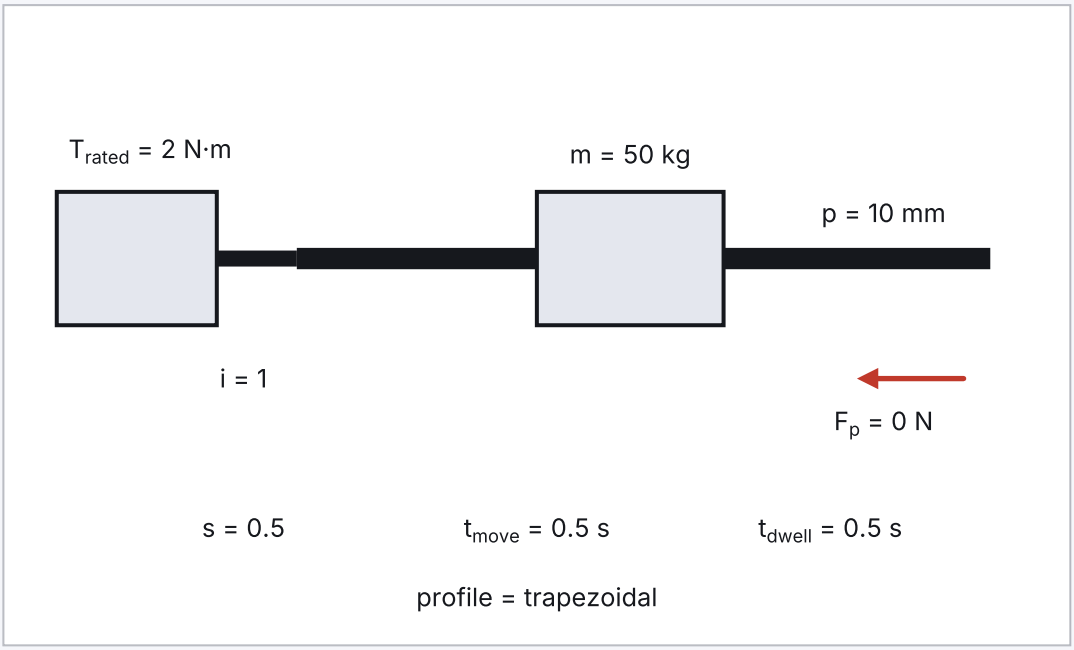


Figure 1 — A motor driving a lead or ball screw; the carriage of mass m moves along the guide. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Linear mechanism	$k = p/(2\pi\, i)$ (screw), $D/(2\, i)$ (belt), $D_p/(2\, i)$ (rack), in metres per motor radian; $J_{refl} = m\, k^2/\eta$; $T_{load} = F\, k/\eta$, with $F = m\, g\, (\sin\theta + \mu \cos\theta) + F_p$
Rotary mechanism through a gearbox	$J_{refl} = J_L/(i^2\, \eta)$; $T_{load} = T_L/(i\, \eta)$
Move profile	trapezoidal $v_{pk} = s/(t_{move} - t_{acc})$, $a = v_{pk}/t_{acc}$; S-curve (sine acceleration, same peak speed) $a_{pk} = (\pi/2)\, a_{trap}$
Peak torque	$T_{acc} = J_{total}\, \alpha$; $T_{pk} = T_{acc} + T_{load}$; $T_{dec} = -T_{acc} + T_{load}$; the check is $SF \cdot \max(\lvert T_{pk}\rvert, \lvert T_{dec}\rvert) \le T_{peak}$; braking power $P_{brake} = \max(0, -T_{dec} \cdot \omega_{pk})$
Thermal duty	$T_{rms} = \sqrt{(\Sigma\, T_i^2\, t_i / t_{cycle})}$ over the whole cycle, the dwell at $T_{hold} = m\, g \sin\theta \cdot k$ for a linear mechanism not held by a brake; $P_{rms} = \omega_{pk}\, \sqrt{\{[t_{acc} \cdot (2/3)(T_{load}^2 + T_{acc}^2) + T_{load}^2\, t_{const}]/t_{cycle}\}}$, and for the S-curve the acceleration term is $t_{acc} \cdot [(3/4)\, T_{load}^2 + (5/16)\, T_{acc}^2]$

WHAT IT CALCULATES	RELATION
Output side and design torque	$T_{alt} = (T_{pk} - T_{dec})/2$ (0 in continuous duty); $T_{out} = T \cdot i \cdot \eta$ and $n_{out} = n_{pk}/i$ for a rotary mechanism; $F_{ax,pk} = F_{load} + m\ a_{pk}$ for a ball screw; T_{design} on the basis T_{basis} (load peak $\max(\lvert T_{pk}\rvert, \lvert T_{dec}\rvert)$, T_{peak}, $SF_c \cdot T_{rated}$ or T_{lim})
Continuous rotary duty	$\omega = 2\pi\ n_L\ i/60$, $n_{pk} = n_L\ i$; $T_{load} = T_L/(i\ \eta)$, $T_{rms} = T_{load}$; $T_{pk} = J_{total}\ \omega/t_{start} + T_{load}$; $P_{cont} = T_{load}\ \omega$, $P_{rms} = P_{cont}$
Rated power of the candidate motor	$P_{rated} = T_{rated} \cdot 2\pi\ n_{rated}/60$ (without n_{rated} there is no P_{rated}); the check is $P_{cont} \le P_{rated}$, $u_{power} = P_{cont}/P_{rated}$

Basis: Load torque, inertia referred to the motor, duty cycles[1](Ch. on motor and drive selection, 5th ed. (edition to be checked)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mechanism	Mechanism	`lead / ball screw`		assumption: Sizing convention.	any
$duty_{mode}$	Duty (point-to-point move cycle, or continuous rotary running with a run-up)	`point-to-point`		default	any
profile	Motion profile	`trapezoidal`		assumption: Sizing convention.	any
T_{basis}	Design torque basis for downstream couplings (T_{design}): the load's peak (the larger of the accelerating and the braking torque, in magnitude), the motor's typed peak torque, a service factor on the rated torque, or a protective limit such as a torque limiter	`Load peak (T_pk)`		default	any
brake	A brake or a self-locking drive holds a linear axis during the dwell (no holding torque); unticked, a vertical or inclined axis is held by the motor	no		default	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
J_{mech}	Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)	0.0001	kg·m²	spec: Screw datasheet BS-1610 + coupling: 1.0e-4 kg·m²	0 … 10 kg·m²
p_{screw}	Screw lead (per revolution of the screw)	10	mm	spec: Screw datasheet BS-1610: 10 mm lead	1 … 100 mm
θ	Incline from horizontal (+ moving uphill)	0	°	assumption: Sizing convention.	-60 … 90 °
F_p	Process force opposing the move (linear)	0	N	assumption: Sizing convention.	0 … 1000000 N
T_L	Constant load torque at the output (rotary; continuous duty needs it, and a zero value is refused)	0	N·m	default	0 … 100000 N·m
m	Moved mass (linear mechanisms)	50	kg	spec: Mechanical layout ML-12: carriage + gripper 50 kg	0 … 100000 kg
i	Gear ratio (motor : output, 1 = none)	1		assumption: Sizing convention.	1 … 1000
η	Mechanism efficiency (overall)	0.9		spec: Screw datasheet BS-1610: 90 %	0.3 … 1
μ	Friction coefficient (guide / bearing)	0.1		assumption: Linear guides with seals: μ = 0.1.	0 … 1
SF	Safety factor on torque	1.25		assumption: Sizing convention.	1 … 3
$J_{ratio,max}$	Inertia ratio limit (load : motor), warning	10		assumption: Sizing convention.	1 … 100
s	Move distance (linear: m; rotary: output revolutions)	0.5		spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.0001 … 10000
t_{move}	Move time	0.5	s	spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.01 … 3600 s

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_{acc}	Acceleration (and deceleration) fraction of the move time	0.333333		spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.05 … 0.5
t_{dwell}	Dwell between moves	0.5	s	spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0 … 3600 s
J_{motor}	Candidate motor rotor inertia	0.0002	kg·m²	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	1e-7 … 10 kg·m²
T_{rated}	Candidate motor continuous (rated) torque	2	N·m	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	0.01 … 100000 N·m
T_{peak}	Candidate motor peak torque	6	N·m	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	0.01 … 100000 N·m
n_{max}	Candidate motor maximum speed	10000	rpm	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	10 … 100000 rpm

3. Assumptions

A1. The transmission is rigid. (Servo-sizing practice.)
A2. The mechanism efficiency applies to the driving direction and is also used in the deceleration phase. (Servo-sizing practice.)
A3. This is conservative for the peak and slightly pessimistic for the RMS torque. (Servo-sizing practice.)
A4. The friction force is Coulomb friction on the moved mass. (Load model.)
A5. Viscous and breakaway friction are not modelled unless included in F_p. (Load model.)
A6. In continuous duty the load torque is constant at the stated speed. (Run-up idealisation for pumps and fans.)
A7. The motor accelerates the whole inertia linearly to speed in t_{start} while carrying the full load torque, which is conservative. (Run-up idealisation for pumps and fans.)
A8. Continuous duty ignores the move distance, move time, acceleration fraction, dwell and profile. (Run-up idealisation for pumps and fans.)
A9. The S-curve profile has sine-shaped acceleration over the same acceleration time. (Jerk-limited profile idealisation.)
A10. The peak acceleration of the S-curve profile is π/2 times the trapezoidal value. (Jerk-limited profile idealisation.)
A11. mechanism (Mechanism) is taken as lead / ball screw. (Sizing convention.)
A12. profile (Motion profile) is taken as trapezoidal. (Sizing convention.)
A13. θ (Incline from horizontal (+ moving uphill)) is taken as 0 °. (Sizing convention.)
A14. F_p (Process force opposing the move (linear)) is taken as 0 N. (Sizing convention.)
A15. i (Gear ratio (motor : output, 1 = none)) is taken as 1. (Sizing convention.)
A16. μ (Friction coefficient (guide / bearing)) is taken as 0.1. (Linear guides with seals: μ = 0.1.)

A17. SF (Safety factor on torque) is taken as 1.25. (Sizing convention.)
A18. $J_{ratio,max}$ (Inertia ratio limit (load : motor), warning) is taken as 10. (Sizing convention.)

4. Working

[22] k = p_{screw} / (2 * pi * i)
= 10 mm / (2 × π × 1)
= 1.5915 mm
Travel per motor radian

[23] F_{load} = m * 9.80665 m/s^2 * (sin(θ) + μ * cos(θ)) + F_p
= 50 kg × 9.80665 m/s^2 × (sin(0 deg) + 0.1 × cos(0 deg)) + 0 N
= 49.033 N
Load force along the motion (gravity, friction, process)

[25] T_{hold} = m * 9.80665 m/s^2 * sin(θ) * k
= 50 kg × 9.80665 m/s^2 × sin(0 deg) × 1.5915 mm
= 0 N·m
Holding torque in the dwell: the motor holds the weight of the axis (efficiency not credited)

[26] T_{load} = F_{load} * k / η
= 49.033 N × 1.5915 mm / 0.9
= 0.08671 N·m
Load torque at the motor shaft

[27] J_{refl} = m * k^2 / η
= 50 kg × (1.5915 mm)^2 / 0.9
= 0.00014072 kg·m²
Load inertia reflected to the motor shaft

[28] v_{pk} = s * 1 m / (t_{move} * (1 - f_{acc}))
= 0.5 × 1 m / (0.5 s × (1 - 0.33333))
= 1.5 m/s
Peak linear velocity (trapezoid)

[29] a_{trap} = v_{pk} / (f_{acc} * t_{move})
= 1.5 m/s / (0.33333 × 0.5 s)
= 9 m/s²
Trapezoidal acceleration

[30] $ω_{pk}$ = v_{pk} / k
= 1.5 m/s / 1.5915 mm
= 942.48 rad/s
Peak motor speed

[32] a_{pk} = a_{trap}
= 9 m/s²
= 9 m/s²
Peak acceleration

[33] $α_{pk}$ = a_{pk} / k
= 9 m/s² / 1.5915 mm
= 5654.9 rad/s²
Peak motor angular acceleration

[34] J_{total} = J_{motor} + J_{mech} + J_{refl}
= 0.0002 kg·m² + 0.0001 kg·m² + 0.00014072 kg·m²
= 0.00044072 kg·m²
Total inertia at the motor shaft

[35] T_{acc} = J_{total} * $α_{pk}$
= 0.00044072 kg·m² × 5654.9 rad/s²
= 2.4922 N·m
Acceleration torque

[36] $T_{pk} = T_{acc} + T_{load}$
$= 2.4922\ N·m + 0.08671\ N·m$
$= 2.5789\ N·m$
Peak torque (accelerating against the load)
[37] $T_{dec} = -T_{acc} + T_{load}$
$= -2.4922\ N·m + 0.08671\ N·m$
$= -2.4055\ N·m$
Torque during deceleration
[38] $T_{alt} = (T_{pk} - T_{dec}) / 2$
$= (2.5789\ N·m - (-2.4055\ N·m)) / 2$
$= 2.4922\ N·m$
Alternating torque amplitude of the cycle about T_{load}
[40] $T_{design} = \max(\text{abs}(T_{pk}), \text{abs}(T_{dec}))$
$= \max(\text{abs}(2.5789\ N·m), \text{abs}(-2.4055\ N·m))$
$= 2.5789\ N·m$
Design torque: the load's peak torque, the larger in magnitude of the accelerating T_{pk} and the braking T_{dec} (an overhauling drive brakes harder than it accelerates)
[41] $F_{ax,pk} = F_{load} + m * a_{pk}$
$= 49.033\ N + 50\ kg \times 9\ m/s^2$
$= 499.03\ N$
Peak axial thrust on the screw's support bearings (load force plus the moved mass accelerated)
[42] $n_{pk} = \omega_{pk}$
$= 942.48\ rad/s$
$= 9000\ rpm$
Peak motor speed
[43] $P_{pk} = T_{pk} * \omega_{pk}$
$= 2.5789\ N·m \times 942.48\ rad/s$
$= 2.4306\ kW$
Peak mechanical power
[44] $P_{brake} = \max(0\ N*m, -T_{dec}) * \omega_{pk}$
$= \max(0\ N*m, --2.4055\ N·m) \times 942.48\ rad/s$
$= 2.2672\ kW$
Peak braking power at the start of the deceleration
[45] $t_{acc} = f_{acc} * t_{move}$
$= 0.33333 \times 0.5\ s$
$= 0.16667\ s$
Acceleration time
[46] $t_{const} = t_{move} - 2 * t_{acc}$
$= 0.5\ s - 2 \times 0.16667\ s$
$= 0.16667\ s$
Constant-velocity time
[47] $t_{cycle} = t_{move} + t_{dwell}$
$= 0.5\ s + 0.5\ s$
$= 1\ s$
Cycle time
[48] $n_{mean} = \omega_{pk} * (t_{move} - t_{acc}) / t_{cycle}$
$= 942.48\ rad/s \times (0.5\ s - 0.16667\ s) / 1\ s$
$= 3000\ rpm$
Mean motor speed over the cycle, dwell included (the travel's motor turns per cycle time)
[49] $P_{rms} = \omega_{pk} * \text{sqrt}((t_{acc} * (2 / 3) * (T_{load}\text{^}2 + T_{acc}\text{^}2) + T_{load}\text{^}2 * t_{const}) / t_{cycle})$
$= 942.48\ rad/s \times \text{sqrt}((0.16667\ s \times 2 / 3 \times ((0.08671\ N·m)\text{^}2 + (2.4922\ N·m)\text{^}2) + (0.08671\ N·m)\text{^}2 \times 0.16667\ s) / 1\ s)$
$= 0.78414\ kW$
RMS mechanical power over the cycle (thermal-equivalent continuous power)

[50] T_{rms} = sqrt((T_{pk}^2 * t_{acc} + T_{load}^2 * t_{const} + T_{dec}^2 * t_{acc} + T_{hold}^2 * t_{dwell}) / t_{cycle})
= sqrt(((2.5789 N·m)^2 × 0.16667 s + (0.08671 N·m)^2 × 0.16667 s + (-2.4055 N·m)^2 × 0.16667 s + (0 N·m)^2 × 0.5 s) / 1 s)
= 1.4402 N·m

RMS torque over the cycle (thermal), the dwell at the holding torque

[51] J_{ratio} = (J_{refl} + J_{mech}) / J_{motor}
= (0.00014072 kg·m² + 0.0001 kg·m²) / 0.0002 kg·m²
= 1.2036

Inertia ratio (load : motor)

[52] $T_{pk,sf}$ = SF * max(abs(T_{pk}), abs(T_{dec}))
= 1.25 × max(abs(2.5789 N·m), abs(-2.4055 N·m))
= 3.2237 N·m

Peak torque with the safety factor, on the larger of the accelerating and the braking torque

[53] $T_{rms,sf}$ = SF * T_{rms}
= 1.25 × 1.4402 N·m
= 1.8002 N·m

RMS torque with the safety factor

[54] u_{peak} = SF * max(abs(T_{pk}), abs(T_{dec})) / T_{peak}
= 1.25 × max(abs(2.5789 N·m), abs(-2.4055 N·m)) / 6 N·m
= 0.53728

Peak torque utilisation with safety factor, on the larger of the accelerating and the braking torque

[55] u_{rms} = SF * T_{rms} / T_{rated}
= 1.25 × 1.4402 N·m / 2 N·m
= 0.90012

Continuous torque utilisation with safety factor

[56] u_{speed} = n_{pk} / n_{max}
= 9000 rpm / 10000 rpm
= 0.9

Speed utilisation

[57] duty = T_{rms} / T_{rated}
= 1.4402 N·m / 2 N·m
= 0.7201

Thermal duty (without margin)

5. Results

SYMBOL	VALUE	DESCRIPTION
T_{load}	0.08671 N·m	Load torque at motor
J_{refl}	0.0001407 kg·m²	Reflected load inertia
J_{total}	0.0004407 kg·m²	Total inertia
α_{pk}	5654.9 rad/s²	Motor angular acceleration
T_{acc}	2.492 N·m	Acceleration torque
T_{pk}	2.579 N·m	Peak torque
T_{dec}	-2.406 N·m	Deceleration torque
T_{rms}	1.440 N·m	RMS torque
T_{design}	2.579 N·m	Design torque for downstream couplings
T_{hold}	0 N·m	Holding torque in the dwell
T_{alt}	2.492 N·m	Alternating torque amplitude

SYMBOL	VALUE	DESCRIPTION
$F_{ax,pk}$	499.0 N	Peak axial thrust on a screw's bearings
n_{pk}	9000 rpm	Peak speed
n_{mean}	3000 rpm	Mean motor speed over the cycle
P_{pk}	2.431 kW	Peak power
P_{brake}	2.267 kW	Peak braking power
P_{rms}	0.7841 kW	RMS power (continuous equivalent)
J_{ratio}	1.2036	Inertia ratio
duty	0.7201	Thermal duty
u_{peak}	0.54	Peak utilisation
u_{rms}	0.90	Continuous utilisation (governing)
u_{speed}	0.90	Speed utilisation
k	1.592 mm	Transmission constant
v_{pk}	1.500 m/s	Peak velocity
a_{pk}	9 m/s²	Peak acceleration

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	RMS torque with the safety factor against the rated torque	1.800 N·m	2 N·m	0.90	
PASS	Peak speed against the maximum speed	9000 rpm	10000 rpm	0.90	
PASS	Peak torque with the safety factor against the motor peak torque	3.224 N·m	6 N·m	0.54	

7. Applicability

WARNING records: tabulated values not taken from a reference record are J_{motor}, T_{rated}, T_{peak}, n_{max} (Candidate motor rating). Enter each as one so its source, edition and licence travel with it. [record]
WARNING n_{pk}: the peak speed is 90 % of n_{max}, and T_{peak} is compared as the stall peak torque. A servo's peak torque falls toward its top speed, so check its torque-speed curve at n_{pk}. [peak-at-speed]
WARNING T_{dec}: the motor brakes during deceleration (negative torque). The drive must absorb the regenerated energy. [overhauling]

Defaults used (inputs left empty):
$duty_{mode}$: not entered, so the default of 'point-to-point' is used.
T_{basis}: not entered, so the default of 'Load peak (T_{pk})' is used.
T_L: not entered, so the default of 0 N·m is used.

8. Verification

Template elec.motor-sizing@1.11.1 Content hash 7ccfbf19d3cec269 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Hughes A, Drury B. Electric Motors and Drives: Fundamentals, Types and Applications. 5th ed. Newnes; 2019.